M.Kumarasamy College of Engineering
NAAC Accredited Autonomous Institution
Approved by AICTE & Affiliated to Anna University
ISO 9001:2015 Certified Institution
Thalavapalayam, Karur - 639 113, TAMILNADU.

A Project Report

on

ONLINE VOTING SYSTEM

Submitted in partial fulfillment of requirements for the award of the course

of

EGB1201 – JAVA PROGRAMMING

Under the guidance of

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Assistant Professor / CSE

Submitted By

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

M.KUMARASAMY COLLEGE OF ENGINEERING
(Autonomous)

KARUR – 639 113

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(Autonomous Institution affiliated to Anna University, Chennai)

KARUR – 639 113

BONAFIDE CERTIFICATE

Certified that this project report on **"ONLINE VOTING SYSTEM"** is the bonafide work of **RITHIK SURIYA M(927623BEC174)** who carried out the project work during the academic year 2024 - 2025 under my supervision.

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

VISION OF THE INSTITUTION

To emerge as a leader among the top institutions in the field of technical education

MISSION OF THE INSTITUTION

- Produce smart technocrats with empirical knowledge who can surmount the global challenges
- Create a diverse, fully-engaged, learner-centric campus environment to provide quality education to the students
- Maintain mutually beneficial partnerships with our alumni, industry, and Professional associations

VISION OF THE DEPARTMENT

To empower the Electronics and Communication Engineering students with emerging technologies, professionalism, innovative research and social responsibility

MISSION OF THE DEPARTMENT

- Attain the academic excellence through innovative teaching learning process, research areas & laboratories and Consultancy projects
- Inculcate the students in problem solving and lifelong learning ability
- Provide entrepreneurial skills and leadership qualities
- Render the technical knowledge and skills of faculty members Produce hi-tech professionals in the field of Electrical and Electronics Engineering by inculcating core knowledge

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1: Core Competence: Graduates will have a successful career in academia or industryassociatedwith Electronics andCommunication Engineering

PEO 2: Professionalism: Graduates will provide feasible solutions for the challenging problemsthroughcomprehensiveresearchandinnovationinthealliedareasofElectronicsandCommunicationEngineering.

PEO 3: Lifelong Learning: Graduates will contribute to the social needs through lifelonglearning,practicingprofessional ethics and leadership quality.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:**Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, give and receive clear instructions.iv

11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Applying knowledge in various areas, like Electronics, Communications, Signal processing, VLSI, Embedded systems etc., in the design and implementation of Engineeringapplication.

PSO2: Able to solve complex problems in Electronics and Communication Engineering with analytical and managerial skills either independently or in team using latest hardware and software tools to fulfil the industrial expectations.

ABSTRACT

- The Online Voting System is a Java-based web application designed for secure and convenient online voting. It ensures secure voter authentication with encrypted login mechanisms to prevent unauthorized access. Voter privacy is maintained as each voter can cast their vote only once, with all votes recorded anonymously. The system uses encrypted data transmission and a robust database for secure storage of voter information and results. Its user-friendly interface simplifies the voting process for users. The backend ensures accurate and secure vote tallying for reliable outcomes. This platform is ideal for small to medium-scale elections, enhancing accuracy, accessibility, and security.

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	POs MAPPED	PSOs MAPPED
The Online Voting System is a Java-based web application designed for secure and convenient online voting. It ensures secure voter authentication with encrypted login mechanisms to prevent unauthorized access. Voter privacy is maintained as each voter can cast their vote only once, with all votes recorded anonymously. The system uses encrypted data transmission and a robust database for secure storage of voter information and results. Its user-friendly interface simplifies the voting process for users. The backend ensures accurate and secure vote tallying for reliable outcomes. This platform is ideal for small to medium-scale elections, enhancing accuracy, accessibility, and security.	**PO1(2)** **PO2(1)** **PO3(2)** **PO4(2)** **PO5(3)** **PO6(2)** **PO7(1)** **PO8(1)** **PO9(3)** **P010(2)** **PO11(2)** **PO12(3)**	**PSO1** **PSO2**

Note: 1- Low, 2-Medium, 3- High

SUPERVISOR **HEAD OF THE DEPARTMENT**

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CHAPTER 1

INTRODUCTION

1.1 Objective

An Online Voting System is a digital platform that enables individuals to cast their votes securely and conveniently over the internet. It is designed to improve voter accessibility, reduce costs, and enhance efficiency in conducting elections. The system typically employs encryption and authentication mechanisms to ensure data privacy and prevent fraud. It streamlines vote counting and provides real-time updates, ensuring transparency in the process. This system is widely used in organizational, governmental, and community-based elections.

1.2 Overview

An Online Voting System is a secure and user-friendly platform that allows individuals to cast their votes electronically over the internet. It aims to simplify the voting process, increase accessibility for remote voters, and reduce manual errors. With features like encryption, authentication, and real-time vote counting, it ensures the integrity and transparency of the election process. The system is adaptable for use in various elections, including government, organizational, and community-based polls. By leveraging technology, it enhances efficiency, saves time, and promotes wider participation.

1.3 JavaProgramming Concepts

1. Object-Oriented Programming (OOP)

2. Data Encryption
3. Database Management
4. User Authentication
5. Graphical User Interface (GUI)

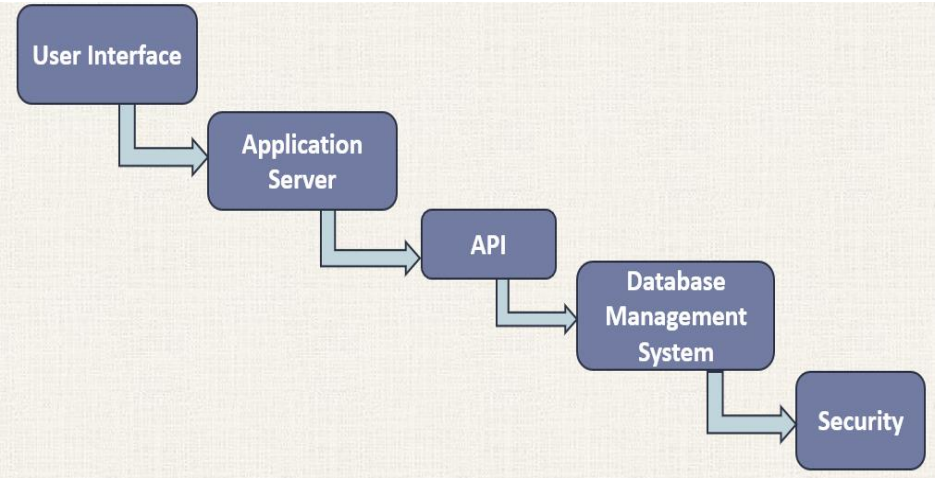
CHAPTER 2

PROJECT METHODOLOGY

2.1Proposed Work

The Online Voting System is a Java-based digital solution designed to streamline the election process by allowing voters to cast ballots online securely and efficiently. It replaces traditional paper-based methods, offering enhanced accessibility for users regardless of location, making it ideal for organizations, schools, or communities with dispersed members. The system ensures robust security through voter authentication, encrypted communication, and data privacy measures, allowing each voter to cast only one vote while keeping votes anonymous. By addressing common issues like manual errors and logistical complications, the platform provides a user-friendly interface for easy navigation and a secure backend for managing voter data and results. Designed for small to medium-scale elections, this system offers a scalable, reliable approach to conducting elections in the digital era.

2.2 Block Diagram


User Interface
Application Server
API
Database Management System
Security

CHAPTER 3

MODULE DESCRIPTION

3.1UserRegistrationModule:

This module allows users to securely register with personal details and unique identifiers. It validates the information and stores it for authentication purposes. The system ensures that only legitimate users can access the platform.

3.2LoginandAuthenticationModule:

This module enables users to log in using encrypted credentials. It authenticates user identities to prevent unauthorized access. Ensures that only registered voters can participate in the election.

3.3ElectionManagementModule:

Administrators can create, manage, and schedule elections. The module allows defining election parameters and candidate details. It securely stores all election configurations and data.

3.4VotingModule:

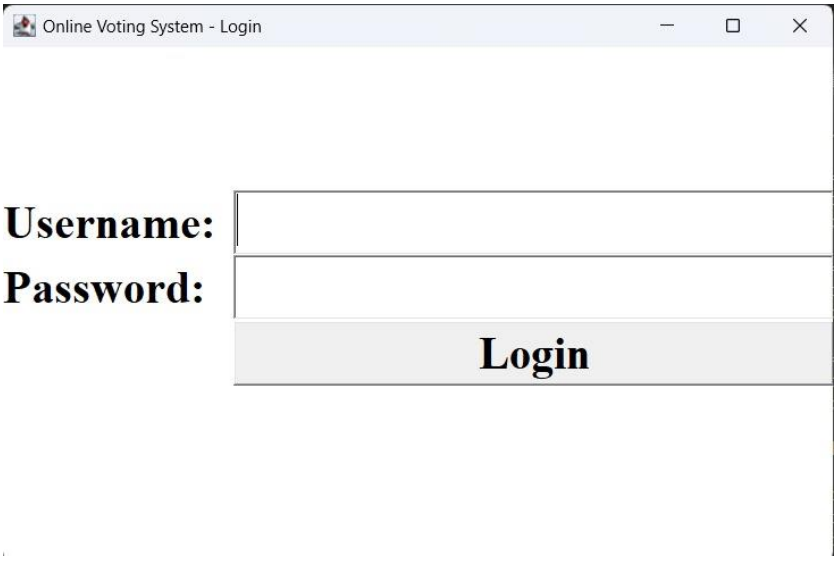
Voters are provided with a simple and intuitive interface to cast their votes. It ensures each user can vote only once, preventing multiple submissions. The votes are recorded anonymously to maintain privacy.

3.5ResultsTallyingandDisplayModule:

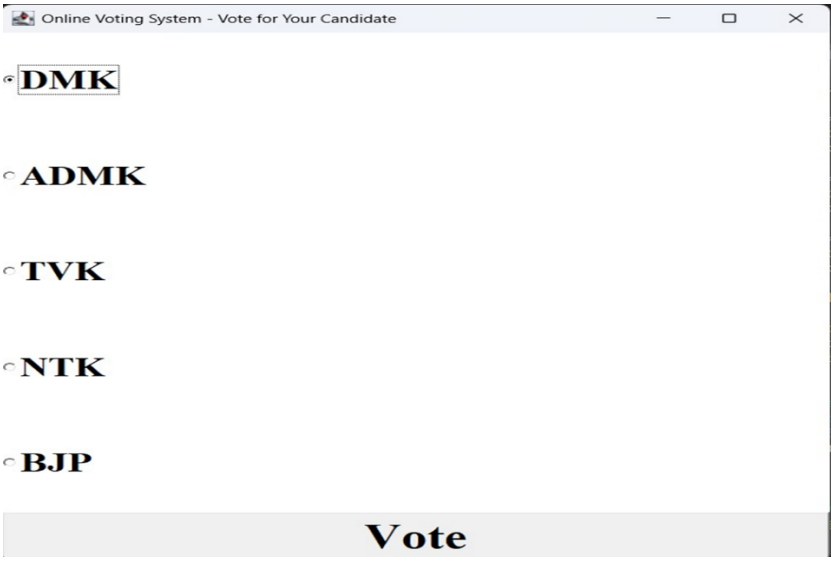
This module automatically tallies votes in real time, ensuring accurate vote counting. It displays the election results clearly for administrators and participants. The results are updated securely and efficiently as votes are cast.

CHAPTER 4

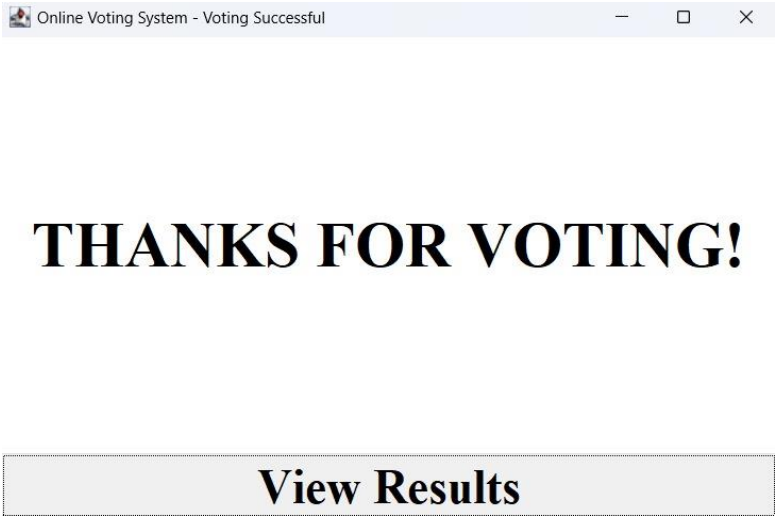
RESULTS AND DISCUSSION



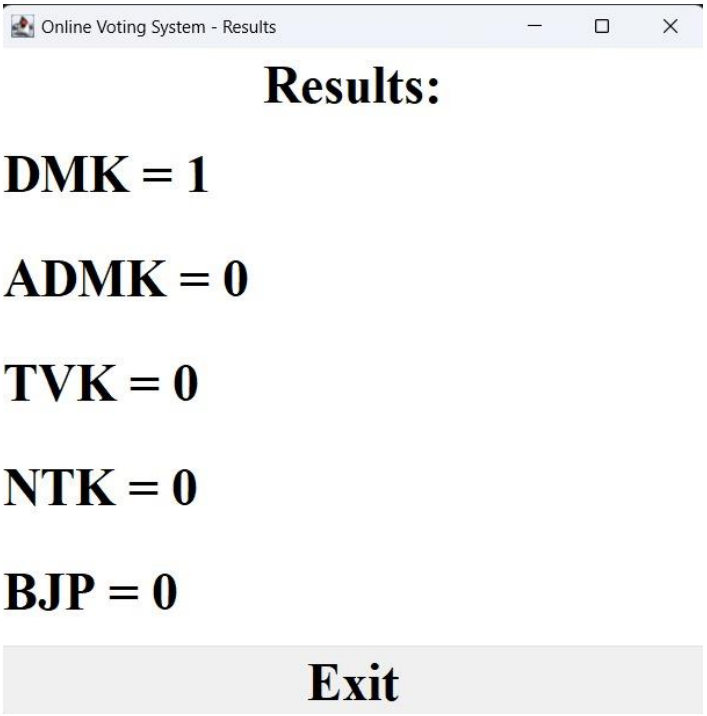
4.1 login interface



4.2 vote selection



4.3 welcome page



4.4 Result page

- The implementation of the Online Voting System demonstrated a successful and secure authentication mechanism. Users were able to log in with valid credentials and cast their votes efficiently, ensuring each voter could only vote once.
- The system's design focused on accessibility and security, incorporating encrypted data transmission and a user-friendly interface. The voting process was seamless, and vote tallying accurately reflected the final results, validating the system's reliability for small to medium-scale elections.
- The Online Voting System successfully authenticated users, enabled secure voting, and accurately tallied results, ensuring a seamless process. Its scalable design and reliable performance make it suitable for small to medium-scale elections.

CHAPTER 5

CONCLUSION

- The Online Voting System offers a secure and efficient solution for small-scale elections, ensuring proper user authentication and vote anonymity. It accurately records and tallies votes, providing a transparent election process. While the system is easy to use, there's room for improvement in the user interface design to make it more visually appealing. Future enhancements could include additional security features and scalability for larger elections. Overall, it delivers a reliable and accessible digital voting platform that can be expanded to meet growing needs.

APPENDIX

(Coding)

```
package Rithik;
import java.awt.*;
import java.awt.event.*;

public class OnlineVotingSystem extends Frame implements ActionListener {
    // Variables for login
    TextField nameField, passwordField;
    Label loginMessage;

    // Variables for voting
    CheckboxGroup candidates;
    Button voteButton, resultsButton;
    int votesForDMK = 0, votesForADMK = 0, votesForTVK = 0, votesForNTK = 0,
votesForBJP = 0;

    // Candidate names
    String candidateAName = "DMK";
    String candidateBName = "ADMK";
    String candidateCName = "TVK";
    String candidateDName = "NTK";
    String candidateEName = "BJP";

    // Constructor to setup login screen
    public OnlineVotingSystem() {
        setLayout(new FlowLayout());

        Label nameLabel = new Label("Username:");
        nameLabel.setFont(new Font("Serif", Font.BOLD, 32));
        nameField = new TextField(20);
        nameField.setFont(new Font("Serif", Font.PLAIN, 32));

        Label passwordLabel = new Label("Password:");
        passwordLabel.setFont(new Font("Serif", Font.BOLD, 32));
        passwordField = new TextField(20);
        passwordField.setFont(new Font("Serif", Font.PLAIN, 32));
        passwordField.setEchoChar('*');
        loginMessage = new Label("");
```

```
    Button loginButton = new Button("Login");
    loginButton.setFont(new Font("Serif", Font.BOLD, 32));
    loginButton.addActionListener(this);

    // Arrange components in separate lines
    setLayout(new GridBagLayout());
    GridBagConstraints gbc = new GridBagConstraints();
    gbc.fill = GridBagConstraints.HORIZONTAL;

    gbc.gridx = 0;
    gbc.gridy = 0;
    add(nameLabel, gbc);

    gbc.gridx = 1;
    gbc.gridy = 0;
    add(nameField, gbc);

    gbc.gridx = 0;
    gbc.gridy = 1;
    add(passwordLabel, gbc);

    gbc.gridx = 1;
    gbc.gridy = 1;
    add(passwordField, gbc);

    gbc.gridx = 1;
    gbc.gridy = 2;
    gbc.anchor = GridBagConstraints.SOUTHEAST;
    add(loginButton, gbc);

    gbc.gridx = 1;
    gbc.gridy = 3;
    add(loginMessage, gbc);

    setTitle("Online Voting System - Login");
    setBackground(Color.WHITE); // Set background to white
    setSize(600, 400);
    setVisible(true);
}

// Action listener for buttons
public void actionPerformed(ActionEvent e) {
```

```
    String command = e.getActionCommand();

    if (command.equals("Login")) {
        String username = nameField.getText();
        String password = passwordField.getText();

        if (username.equals("admin") && password.equals("123")) {
            // Clear the login screen
            removeAll();
            setupVotingPage();
        } else {
            loginMessage.setText("Invalid login. Try again.");
        }
    } else if (command.equals("Vote")) {
        Checkbox selectedCandidate = candidates.getSelectedCheckbox();

        if (selectedCandidate != null) {
            if (selectedCandidate.getLabel().equals(candidateAName)) {
                votesForDMK++;
            } else if (selectedCandidate.getLabel().equals(candidateBName)) {
                votesForADMK++;
            } else if (selectedCandidate.getLabel().equals(candidateCName)) {
                votesForTVK++;
            } else if (selectedCandidate.getLabel().equals(candidateDName)) {
                votesForNTK++;
            } else if (selectedCandidate.getLabel().equals(candidateEName)) {
                votesForBJP++;
            }
            showVotingSuccessful();
        } else {
            loginMessage.setText("Please select a candidate.");
        }
    } else if (command.equals("View Results")) {
        showResults();
    }
}

// Setup the voting screen
private void setupVotingPage() {
    setLayout(new BorderLayout());

    candidates = new CheckboxGroup();
```

```
    Panel candidatesPanel = new Panel(new GridLayout(5, 1));
    candidatesPanel.setBackground(Color.WHITE); // Set background to white

    // Use bold and larger font size for candidates
    Font candidateFont = new Font("Serif", Font.BOLD, 28);

    Checkbox candidateA = new Checkbox(candidateAName, candidates, false);
    candidateA.setFont(candidateFont);
    candidatesPanel.add(candidateA);

    Checkbox candidateB = new Checkbox(candidateBName, candidates, false);
    candidateB.setFont(candidateFont);
    candidatesPanel.add(candidateB);

    Checkbox candidateC = new Checkbox(candidateCName, candidates, false);
    candidateC.setFont(candidateFont);
    candidatesPanel.add(candidateC);

    Checkbox candidateD = new Checkbox(candidateDName, candidates, false);
    candidateD.setFont(candidateFont);
    candidatesPanel.add(candidateD);

    Checkbox candidateE = new Checkbox(candidateEName, candidates, false);
    candidateE.setFont(candidateFont);
    candidatesPanel.add(candidateE);

    voteButton = new Button("Vote");
    voteButton.setFont(new Font("Serif", Font.BOLD, 36));
    voteButton.addActionListener(this);

    add(candidatesPanel, BorderLayout.CENTER);
    add(voteButton, BorderLayout.SOUTH);

    setTitle("Online Voting System - Vote for Your Candidate");
    setBackground(Color.WHITE); // Set background to white
    setSize(600, 600);
    setVisible(true);
}

// Show voting successful screen
private void showVotingSuccessful() {
    removeAll();
```

```
        setLayout(new BorderLayout());

        Label successMessage = new Label("THANKS FOR VOTING!",
Label.CENTER);
        successMessage.setFont(new Font("Serif", Font.BOLD, 48));
        successMessage.setBackground(Color.WHITE); // Set background to white

        resultsButton = new Button("View Results");
        resultsButton.setFont(new Font("Serif", Font.BOLD, 36));
        resultsButton.addActionListener(this);

        add(successMessage, BorderLayout.CENTER);
        add(resultsButton, BorderLayout.SOUTH);

        setTitle("Online Voting System - Voting Successful");
        setBackground(Color.WHITE); // Set background to white
        setSize(600, 400);
        setVisible(true);
    }

    // Show results screen with updated size
    private void showResults() {
        removeAll();
        setLayout(new BorderLayout());

        Label resultsLabel = new Label("Results:", Label.CENTER);
        resultsLabel.setFont(new Font("Serif", Font.BOLD, 36));
        resultsLabel.setBackground(Color.WHITE); // Set background to white
        add(resultsLabel, BorderLayout.NORTH);

        Panel resultsPanel = new Panel(new GridLayout(5, 1));
        resultsPanel.setBackground(Color.WHITE); // Set background to white

        Font resultsFont = new Font("Serif", Font.BOLD, 36);

        Label dmkLabel = new Label(candidateAName + " = " + votesForDMK);
        dmkLabel.setFont(resultsFont);
        resultsPanel.add(dmkLabel);

        Label admkLabel = new Label(candidateBName + " = " + votesForADMK);
        admkLabel.setFont(resultsFont);
        resultsPanel.add(admkLabel);
```

```
        Label tvkLabel = new Label(candidateCName + " = " + votesForTVK);
        tvkLabel.setFont(resultsFont);
        resultsPanel.add(tvkLabel);

        Label ntkLabel = new Label(candidateDName + " = " + votesForNTK);
        ntkLabel.setFont(resultsFont);
        resultsPanel.add(ntkLabel);

        Label bjpLabel = new Label(candidateEName + " = " + votesForBJP);
        bjpLabel.setFont(resultsFont);
        resultsPanel.add(bjpLabel);

        add(resultsPanel, BorderLayout.CENTER);

        Button exitButton = new Button("Exit");
        exitButton.setFont(new Font("Serif", Font.BOLD, 36));
        exitButton.addActionListener(e -> System.exit(0));
        add(exitButton, BorderLayout.SOUTH);

        setTitle("Online Voting System - Results");
        setBackground(Color.WHITE); // Set background to white
        setSize(500, 500);  // Adjusted the results page size to 800x600
        setVisible(true);
    }

    // Main method
    public static void main(String[] args) {
        new OnlineVotingSystem();
    }
}
```